

Sophia Huang

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EDUCATION

Binghamton University, State University of New York

Master of Science in Data Analytics – Watson College of Engineering and Applied Science

Expected June 2026

Cumulative GPA: 3.93/4.00

Relevant Courses: Machine Learning & Data Mining, AI Driven Decision Support, Analytics for Enterprise Apps

Bachelor of Arts in Psychology, Minor in Digital and Data Studies – Harpur College

May 2025

Cumulative GPA: 3.86/4.00 | Dean's List Spring 2022 – Spring 2026

Instituto Lorenzo de' Medici, Florence, Italy

May 2024

Italian Study Abroad Program, Gilman Scholarship Recipient

TECHNICAL SKILLS

Programming & Data Analysis: Python, R, SQL, Bash

Libraries & Tools: PyTorch, Pandas, NumPy, TensorFlow, Scikit-learn, Fastai, Shiny, AWS (Lambda, EC2, DynamoDB, S3), Linux, Figma, Redshift, Neo4j, MongoDB

Visualization & Analysis: Tableau, Power BI, Excel

Statistical Concepts: Hypothesis Testing, Regression Analysis, ANOVA, Probability Distribution, Time Series Analysis

Other: 2nd Degree Black Belt in Karate, Mandarin (Fluent)

PROFESSIONAL EXPERIENCE

Cognition, Aging, and Memory Performance (CAMP) Lab, Binghamton University, Vestal, NY

Research Data Scientist

October 2025 – May 2026

- Engineered an automated Python pipeline using pandas and NumPy to transform raw eye tracking data files into analysis-ready data for 100+ participants, reducing errors by 90% and processing time from 3-4 hours to seconds
- Developed entropy metric calculations by implementing reusable functions applied across fields and participants, producing comparable features for statistical modeling and reporting on 400+ research variables

Information Technology Services, Binghamton University, Vestal, NY

Lead Student Help Desk Analyst

August 2023 – May 2026

- Promoted to lead analyst after 2 years, supervising and onboarding 11 student workers and coordinating schedules to ensure seamless workflow and service delivery across a team supporting 100+ users
- Achieved a 90% first-contact resolution rate across 300+ annual support tickets by implementing streamlined troubleshooting protocols, minimizing escalations and optimizing support efficiency
- Diagnosed and resolved software and hardware failures for 50+ Mac and Windows machines, applying root cause analysis to restore functionality for screens, keyboards, and system errors
- Received positive feedback from over 95% of campus community members supported via phone, email, and in-person interactions, translating complex technical concepts into clear, accessible explanations

PROJECT EXPERIENCE

Campus Parking Availability Detection

January 2026 – May 2026

- Built a computer vision pipeline using YOLO26 to detect and classify parking space occupancy from lot images, archiving 90% precision and 93% recall on a 6,000+ image test set of 255,000+ labeled spaces
- Designed an interactive mobile application prototyped in Figma featuring real-time color-coded lot maps, section breakdowns, peak hours charts, and day/time filtering to help commuters make informed parking decisions
- Established a full scalability framework for deploying the model on Binghamton University's existing camera infrastructure with minimal hardware investment

United Health Services (UHS) 30-Day Hospital Readmission Analysis

January 2026 – May 2026

- Analyzed 5,200+ inpatient hospitalizations from UHS Epic EHR data to identify key drivers of 30-day readmission and evaluate the effectiveness of three care transition interventions across CMS-tracked conditions
- Engineered features including TCM follow-up time windows, a composite transition bundle score, and SDOH burden and clustering variables using K-Modes on 13 binary social determinant flags
- Applied inverse probability weighting to balance treated/untreated groups, reducing SMDs from 0.15 to 0.05 (risk score) and 0.20 to 0.004 (SDOH score), supporting credible causal inference from observational data
- Delivered actionable recommendations to UHS, identifying SDOH factors (transportation, food insecurity, housing instability) as high-priority pre-discharge intervention targets

INTERESTS

Photography - Karate - Boxing - Rock Climbing - Snowboarding – Traveling